Aspect-Based Sentiment Analysis on Hotel Reviews

UE24CS352A Machine Learning | Team 22 | Project 1
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Objective and data

Predict overall and aspect-specific ratings (1-5), then map 1-2 to negative, 3 to neutral and 4-5 to positive. The upgrade investigates whether clause routing reduces cross-aspect contamination, such as positive location language masking a dirty room. The primary evaluation uses **2,100 real reviews from 122 hotels** in the ArguAna TripAdvisor annotated v2 corpus [1]. Title/review text comes from XMI Sofa text, overall from Sentiment.score, and aspect ratings from Aspects. Hotel category is never used as review sentiment.

Observed ratings: Overall 2,100; Service 1,608; Rooms 1,647; Location 1,238; Value 1,631; Cleanliness 1,651; Business Service 772. Source -1 stays missing. **Real Sleep Quality remains unavailable.** Author metadata is omitted. Source and CSV checksums are recorded. The separate 320-row synthetic sample is demonstration only; no synthetic labels enter real evaluation. The real corpus permits scientific use, not commercial applications.

Method and leakage protection

A stateless router segments sentences and contrast/independent clauses, preserves negation, handles vocabulary variants, and assigns relevant clauses to one or more aspects. Confident routes use aspect_text; uncertain/no-explicit-mention routes fall back to the full review with a recorded reason. The selected transformer lives inside both training and inference pipelines. Overall always uses full text. Explicit mention rules gate the UI by default, but do not remove labelled test rows from evaluation.

For each target, compare both full_review and aspect_text with three fixed families: TF-IDF + Multinomial NB (alpha 0.5), count features + balanced Linear SVM (C 0.1), and TF-IDF + balanced Logistic Regression (C 4). All use unigrams/bigrams and at most 12,000 features. Vocabulary/IDF fit on training only. Seed 42 preserves **1,342 train / 388 validation / 370 test** reviews with hotels disjoint. Validation polarity macro-F1 selects each configuration, with deterministic ties. Refit on train + validation before final test. The historical test was already known, so this is a descriptive reused-holdout comparison.

NB/LR provide native uncalibrated probabilities. SVM supports out-of-fold sigmoid calibration, grouped by hotel when available [3]. Validation correctness determines confidence thresholds before refit. Low-confidence predictions abstain; candidate ratings remain visible in details. Statuses distinguish predicted, uncertain, not_mentioned, unavailable and unknown vocabulary. No thresholds, vocabulary, calibration or routing rules learn from test data.

Validation selection and test comparison

Target	Selected model / text	Val. F1 old / new	Test F1 old / new
Overall	NB / full	0.666 / 0.666	0.709 / 0.709
Service	LR / full	0.635 / 0.635	0.628 / 0.628
Rooms	LR / full	0.588 / 0.588	0.560 / 0.560
Location	LR / routed	0.411 / 0.413	0.423 / 0.383
Value	NB / full	0.600 / 0.600	0.615 / 0.615
Cleanliness	LR / routed	0.504 / 0.567	0.539 / 0.566
Business	LR / full	0.548 / 0.548	0.558 / 0.558

The new full-review baseline exactly reproduces archived scores on the unchanged split. Routing wins validation for Location and Cleanliness only. No choice was reversed after observing test results.

Results, diagnostics and verification

Real held-out results below score every observed target label before confidence or mention filtering. Macro-F1 gives the three polarity classes equal weight. Majority star and polarity baselines use development labels only.

Target / n	Star acc.	Within 1	MAE	Polarity acc.	Macro-F1	Majority pol.
Overall / 370	51.1%	92.2%	0.608	74.6%	0.709	32.4%
Service / 271	44.6%	84.5%	0.764	68.3%	0.628	45.0%
Rooms / 276	39.1%	80.8%	0.873	60.9%	0.560	47.8%
Location / 223	42.2%	83.0%	0.807	72.2%	0.383	78.9%
Value / 279	46.2%	85.3%	0.710	62.4%	0.615	37.3%
Cleanliness / 278	43.9%	81.7%	0.784	65.1%	0.566	61.5%
Business / 130	38.5%	76.9%	0.877	55.4%	0.558	39.2%

Interpretation and uncertainty

Cleanliness: polarity macro-F1 rises 0.539 to 0.566, while exact-star accuracy falls 46.4% to 43.9% and MAE increases 0.755 to 0.784. **Location:** the small validation gain does not generalize; test accuracy falls 74.4% to 72.2%, below the 78.9% majority baseline. Its 223 test labels include 176 positive, 27 neutral and 20 negative. Negative/neutral recall is only 10.0%/14.8%. **Business Service:** only 499/143/130 labelled rows; unchanged balanced LR yields macro-F1 0.558. Negative/neutral/positive F1 is 0.667/0.495/0.512.

Paired hotel bootstrap (500 seeded replicates) gives a 95% interval for macro-F1 change of [-0.064, +0.083] for Cleanliness and [-0.122, +0.088] for Location. Both include zero. These intervals condition on fitted models and exclude retraining variability. On mixed-label rows, Cleanliness F1 rises 0.511 to 0.560 (n=190), Location falls 0.412 to 0.322 (n=162), and Rooms stays 0.460 (n=189). Disagreeing aspect labels are a descriptive proxy, not annotated sentiment spans.

Demo behavior and remaining errors

In the long mixed review praising location, staff and value but describing an extremely dirty room, routed Cleanliness predicts 1/negative while full-review Rooms remains 4/positive. The router preserves "not clean", but the model abstains with a neutral candidate. "Not bad" can still receive a negative Rooms prediction, and isolated "location was terrible" remains wrongly positive. All nine requested cases and before/after outputs are retained. These are qualitative probes, not an accuracy benchmark.

Confidence improves transparency, not guaranteed correctness. Overall accepts 61.6% of test rows with 88.2% selective polarity accuracy, versus 74.6% on all rows. Business Service emits UI predictions for only 13.8% of labelled test rows, with 88.9% accuracy on those accepted rows. Coverage must accompany selective accuracy. Rule-based mentions miss implicit aspects; native probabilities are uncalibrated and may shift after refit. Historical English data, missing ratings and review-level rather than clause-level supervision limit generalization.

Implementation and testing

78 tests pass with zero failures, errors or skips. Tests cover routing/negation separately from learned sentiment, evidence/fallback, uncertainty, missing aspects, real/synthetic isolation, train-only features, validation selection, hotel-aware calibration, serialization, CLI and Streamlit interactions. The verifier independently recomputes saved predictions/metrics, raw validation thresholds and candidate scores; checks model/source hashes, fitted vocabularies and split/calibration isolation; and starts a real Streamlit server. Dataset and label bytes remain unchanged.

Run **python run_project.py** to train, **python run_project.py --review "The staff were friendly."** for saved-model inference, **python -m streamlit run streamlit_app.py** for the demo and **python verify_project.py** for final readiness evidence. README, DEMO_GUIDE and UPGRADE_REPORT give full commands and limitations. The result is a working university prototype. Transformers and richer labelled aspect extraction remain future work. Private GitHub access and truthful contributions require team confirmation.

References

[1] Wachsmuth et al. (2014), A Review Corpus for Argumentation Analysis. zenodo.org/records/3973241.
[2] Yu (2016), Aspect-based Sentiment Analysis on Hotel Reviews, Stanford CS229. cs229.stanford.edu/proj2016spr/report/032.pdf. Inspiration; our results are not a reproduction.
[3] scikit-learn, Probability calibration. scikit-learn.org/stable/modules/calibration.html.